Question

Element **A** is generally found only in associated minerals, and independent deposits are rare. When a mineral containing **A** is mixed with potassium carbonate and burned in the presence of air, a potassium salt **B** containing **A** is obtained. Upon acidification and evaporative crystallization of **B**, colorless crystals **C** are produced. **C** readily dehydrates to form a yellow solid **D**. When **D** reacts with SF_4 at temperatures above $120\,℃$, only a gas **E** containing **A** with a mass fraction of 62.67% and another gas **F** are obtained. **F** reacts with an aqueous solution of **C** to precipitate black crystals **G**. When **G** reacts with SbF_5 in SO_2 at $-30\,℃$ in a stoichiometric ratio of $8:5$, an ionic compound **H** and SbF_3 are produced in a $1:1$ product ratio. The cation of compound **H** has a symmetric and cyclic structure.

Please analyze the substances described above through reasonable logic or conjecture, determine what each substance is (providing molecular formulas rather than simplest formulas), and answer whether the following options are correct.

A. In an infinitely layered **D**, let the other element besides **A** be denoted as **Y**. Then, the ratio of terminal **Y** to bridging **Y** is $2:1$.

B. The gas **E** has a square planar structure.

C. The element in gas **F** with the highest oxidation state has a structure similar to that of element **G**.

D. **H** is a $1:1$ type ionic compound

E. There is no **A-A** bond in the cation of **H**.

F. It is known that one of the conformations of the **H** cation has two mirror planes, thus its molecular point group is the same as that of the water molecule.

G. None of the other options are correct.

H. G has a crystal structure with $\mathbf{A}_8$ as the structural motif.

Answer

Correct Answer: F

Detailed Explanation

At first glance, this question is difficult to answer directly. Therefore, start with **E**, which has a definite mass fraction, and first determine which other elements **E** contains.

From the question, SF_4 acts as a fluorinating agent here, and undergoes a metathesis-like reaction with the oxide **D** formed after acid dehydration. **F** should be SO_2, then **E** is a fluoride.

CHECKPOINT 1 PTS

SF_4 acts as a fluorinating agent here, **F** is SO_2, **E** is a fluoride.

Substituting the mass fraction, the relative atomic mass of **A** can be calculated. The mass fraction of **A** in

$$AF_4$$

is

$$\frac{x}{x+4\times 19} \approx 45.23\%$$

, solving for $x = 127.6(\mathrm{g/mol})$, corresponding to the element tellurium.

CHECKPOINT 1 PTS

The relative atomic mass of **A** is approximately 127.6, corresponding to the element tellurium.

The answers from **A** to **F** in this question are obvious.

Under the action of alkali, tellurium can be oxidized by oxygen, but it is difficult to be directly oxidized to the highest valence +6, it can only be oxidized to the +4 valence, to obtain potassium tellurite **B**.

CHECKPOINT 1 PTS

In **B**, Te has a +4 valence, i.e., it is K_2TeO_3

Acidification of potassium tellurite yields tellurous acid **C**, dehydration yields tellurium dioxide **D**, and the logic up to this point is very smooth.

CHECKPOINT 1 PTS

C is tellurous acid H_2TeO_3, **D** is tellurium dioxide TeO_2.

Based on this, answer options A and B: Analyzing the atomic ratio, in layered

$$TeO_2$$

, the ratio of tellurium to bridging oxygen is obviously $1:4$, each tellurium corresponds to 4 bridging oxygens, and there are no terminal oxygen atoms, A is incorrect;

CHECKPOINT 1 PTS

Tellurium dioxide has no terminal oxygens

Using VSEPR to judge, in

$$TeF_4$$

, Te has 6 electrons, forms four bonds, and has one lone pair of electrons remaining, which is a distorted tetrahedral structure, B is incorrect.

CHECKPOINT 1 PTS

Tellurium tetrafluoride has a distorted tetrahedral structure

$$SO_2$$

is a reducing agent, and undergoes a redox reaction with the oxidizing agent

$$H_2TeO_3$$

, the product should be elemental tellurium (acid-base reaction may be considered here, but the product cannot obtain a precipitate). Te does not have a $\mathrm{Te_8}$ cyclic structure, but a spiral crystal, H is incorrect

CHECKPOINT 1 PTS

SO_2 acts as a reducing agent here

$$\mathrm{SbF_5}$$

can act as both an oxidizing agent and a coordinating agent. The production of

$$\mathrm{SbF_3}$$

indicates that it does act as an oxidizing agent.

Trying it out, it can be found that

$$\mathrm{SbF_5}$$

cannot avoid acting as a Lewis acid. Discuss the common anions such as

$$[\mathrm{Sb_2F_{11}}]^-$$

or

$$[SbF_6]^-$$

respectively, try to balance the equation, and finally it is concluded that it is the former, and the chemical formula of **H** can be obtained as

$$Te_8[Sb_2F_{11}]_2$$

.

CHECKPOINT 2 PTS

Analyze that **G** is elemental tellurium, and give the information that **H** contains Te_8^{2+}

Based on this, it can be known that the element in the highest oxidation state in the gas **F** is S_8, which has a different structure from the spiral crystal structure of Te, C is incorrect.

H is obviously a $1:2$ type ionic compound, D is incorrect.

The

$$Te_8^{2+}$$

ion combined with the question prompt (symmetric and cyclic structure) should be a fused five-membered ring structure. Combining VSEPR knowledge, Te and its homolog O form two bonds when uncharged; when forming three bonds, it should carry a formal positive charge to satisfy the octet rule, each Te bridging two Te carries one positive charge, and the ion as a whole carries two positive charges. Therefore, there is bonding interaction between Te and Te, E is incorrect.

CHECKPOINT 2 PTS

Te_8^{2+} ion contains $\mathrm{Te-Te}$ bonds

Combining organic chemistry knowledge, there are two structures for the fused five-membered ring, i.e., cis-trans isomers, where the cis structure has one two-fold axis and two mirror planes, C_{2v} point group, which is consistent with the water molecule and meets the requirements of the question, F is correct.

CHECKPOINT 1 PTS

There exists a certain (cis) Te_8^{2+} structure that is consistent with the molecular point group of the water molecule

In summary, the substances are:

- A:

$$\mathrm{Te}$$

(Tellurium, verified by mass fraction calculation)

- **B**:

$$K_2TeO_3$$

(Potassium carbonate + mineral calcined in air, oxidized to potassium tellurite)

- **C**:

$$H_2TeO_3$$

(B acidified to obtain tellurous acid)

- **D**:

$$TeO_2$$

(Tellurous acid dehydrated to obtain tellurium dioxide, yellow solid)

- **E**:

$$TeF_4$$

(

$$TeO_2 + SF_4 \rightarrow TeF_4 + SO_2$$

, distorted tetrahedral structure)

- **F**:

$$SO_2$$

(Byproduct)

- **G**:

$$Te$$

(

$$SO_2$$

reacts with

$$H_2TeO_3$$

to precipitate fused cyclic

$$Te_8$$

, black crystal)

- **H**:

$$Te_8[Sb_2F_{11}]_2$$

(

$$8Te + 5SbF_5 \rightarrow Te_8^{2+} \cdot 2[Sb_2F_{11}]^- + SbF_3)$$

)

The following is a summary of the answers for each option:

A. In the long-chain

$$TeO_2$$

, the ratio of tellurium to bridging oxygen is obviously $1:1$, and each tellurium has one terminal oxygen, so the required ratio is $1:1$, which is incorrect.

B.

$$TeF_4$$

is a distorted tetrahedral structure, which is incorrect.

C. The element in the highest oxidation state in the gas **F** is S_8, which has a similar structure to Te_8, but is yellow in color, the colors are different, which is incorrect.

D. **H** is a $1:2$ type ionic compound, which is incorrect.

E. Te_8^{2+} has a fused five-membered ring structure, and the two bridging Te have bonding interactions, which is incorrect.

F. There are two types of fused five-membered ring structures, cis and trans. The cis structure has one two-fold axis and two mirror planes, C_{2v} point group, which is consistent with the water molecule and meets the requirements of the question, which is correct. Therefore, G is incorrect.

The answer to this question is F.